TEXORA · SESSIONAL NOTES · PRACTICAL STUDY SHEET

Practical Study of Jute Spinning Machines: Drawing Frame and Spinning Frames

The long-staple counterpart of Study 1: how to study a jute drawing frame's pinned faller field — parts to identify, the drafting action to watch, reach and draft to measure — and the flyer spinning frame that turns finisher sliver into jute yarn.

TE-2202 · SPINNING TECHNOLOGY-II SESSIONAL · STUDY 2 · LEVEL 2 TERM 2

1. Object and method of the study

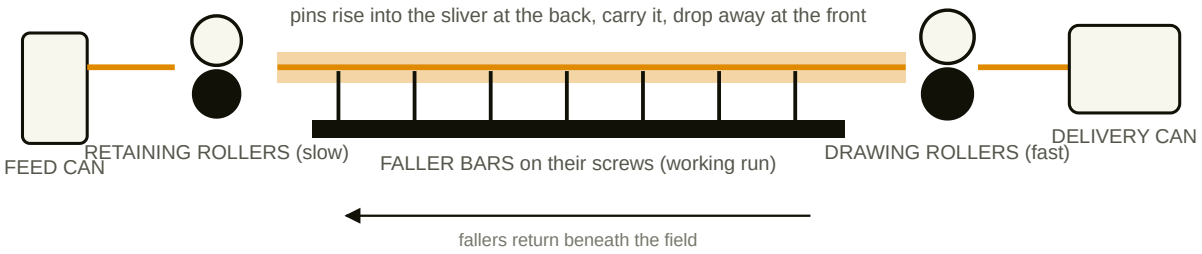
Object: to identify the working parts of the jute drawing frame and the jute spinning (flyer) frame, to observe the faller-bar drafting field in action, and to record the settings — doublings, draft, reach, speeds, twist — that connect the machines to the theory of TE-2201 Topic 7.

Method: the same routine as Study 1 — trace the material path, identify parts, read settings without altering them, record production points, and sketch the drafting field (Figure 1 is your model). Jute machinery runs heavy strands at modest speeds; the faller field can be watched in motion, which is the great advantage of this study: stand where you can see the pins enter the sliver at the back of the field and drop away at the front, and the whole drafting principle of Topic 7 is visible at once.

2. The jute drawing frame — parts to identify

- **Feed cans and feed sheet** — card sliver (first passage) or drawing sliver (later passages) presented to the feed rollers; count the ends up — that is the **doubling** of this passage.
- **Retaining rollers** — the slow roller pair that meters the sliver into the drafting field; their surface speed is the denominator of the draft.
- **Faller bars and gill pins** — the heart of the machine: bars carrying rows of pins, driven forward by **screws** (find the screw shafts under the field), rising into the sliver at the back of the field, travelling with it, and dropping away at the front to return beneath.
- **Drawing rollers** — the fast roller pair at the front of the field; their surface speed, against the retaining speed, sets the draft.

- **Delivery and coiler** — the delivered sliver is condensed and coiled into the delivery can (at the finisher passage, prepared in the form the spinning frame receives).
- **Drive and change wheels** — locate the draft change wheel and the faller-speed drive; note which wheels the lab has fitted.



WATCH: FIBRES SEIZED BY THE DRAWING ROLLERS SLIDE THROUGH THE PINS — THAT SLIDING IS THE DRAFT

Figure 1 — The jute drawing frame's drafting field, section view. Retaining rollers feed the sliver onto a moving bed of faller bars whose pins project up into the strand mass; the fast drawing rollers pull the leading fibres through the pins, and the fallers drop away at the front and return beneath the field on their screws.

3. Drafting-field observations

- **Pin behaviour:** watch one faller bar through its journey — pins enter the sliver cleanly at the back (no fibre dragged under), travel fully loaded, and leave at the front without lifting the sliver. Note any bent, missing or loose pins and what they do to the strand passing over them.
- **Where the draft happens:** observe that fibres move at faller speed until their leading ends reach the drawing rollers, then accelerate — the drafting point is spread through the field, not concentrated at one nip. Compare with the concentrated nip drafting of the cotton system in Study 1.
- **Slip:** at a spinning frame working by **slip draft**, note how much of the strand slides relative to the fallers; at a **controlled-draft** frame, find the extra control (retaining rollers/aprons) that suppresses slip.
- **Strand condition:** record slubs, lumps or clouds in the delivered sliver and relate them to pin condition, doubling number and draft size.

4. Settings to record — drawing frame

ITEM	WHAT TO WRITE IN THE REPORT
Passage	First (breaker), second (inter) or finisher drawing — and the evidence (sliver weight, pin fineness) that tells you which

Doublings	Number of feed ends up at this passage
Draft	Drawing-roller surface speed ÷ retaining-roller surface speed, from the fitted change wheels; check it also equals (feed sliver weight × doublings) ÷ delivered sliver weight
Reach	Distance between retaining nip and drawing nip, against the cut length of the material being run (reach must exceed the longest strands — Topic 7)
Faller field	Faller pitch / pin density (pins per unit length and rows), pin condition, faller speed if displayed
Sliver weights	Feed and delivered sliver weight per unit length as the lab measures them (weigh a cut length); the direct-system basis of jute weighing stated explicitly

DRAFT CHECK — TWO WAYS, ONE ANSWER

Draft from speeds: drawing rollers 16 m/min, retaining rollers 2.0 m/min → draft = 8. Draft from weights: 4 ends of feed sliver at 4 units each = 16 units in, delivered sliver 2 units → draft = 8. If your two figures disagree, your weights or your speed readings are wrong — find out which before leaving the machine; that disagreement is the most instructive observation this study offers.

5. The jute spinning (flyer) frame — parts and observations

5.1 Parts to identify

- **Feed** — finisher sliver cans (or rove bobbins on frames that take a rove) behind the frame; the sliver guide into the retaining rollers.
- **Drafting field** — retaining rollers and the faller field, working by slip draft or controlled draft (identify which, per Section 3).
- **Flyer and spindle** — the large flyer rotating about the bobbin; find the flyer legs and eyes, the spindle, and the bobbin on its rail.
- **Bobbin rail and builder motion** — the traversing rail that lays the coils and shortens its stroke to taper the package ends.
- **Delivery rollers** — the front rollers that meter the drafted strand to the flyer.

5.2 Settings and production points to record

- **Draft** at the spinning frame (typically ≈ 8–15, Topic 7) and the drafting system in use.

- **Flyer (spindle) speed** — expect roughly 3,000–4,500 rpm territory, far below cotton ring-frame speeds; record the actual figure.
- **Twist** — turns per inch from flyer speed ÷ delivery; state whether the yarn being spun is warp or weft and compare with the Topic 7 ranges (warp ≈ 3.2–3.8 tpi, weft ≈ 2.4–3.0 tpi).
- **Yarn count** — in the jute direct system: pounds per spindle of 14,400 yd. Higher number = coarser yarn (the opposite sense to cotton Ne); record the count being spun and weigh a measured length to verify it.
- **Bobbin build** — sketch the package shape, note the traverse shortening, and record package weight at doff.

6. What the lab report must contain

SECTION	CONTENT
Machine identity	Maker/model from the plate; passage (drawing) or frame type (spinning); material and counts being processed
Material path	Traced path with a labelled sketch of the drafting field (Figure 1 as model)
Parts table	Each part — retaining rollers, fallers, screws, drawing rollers, flyer, rail — with its function in one line
Settings record	The tables of Section 4 and Section 5.2 filled with observed values
Draft verification	The two-way draft check of the keybox, with your figures and any discrepancy explained
Faults observed	Pin damage, slip irregularity, slubs — each with its probable cause and remedy from Topic 7

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why can a jute sliver not be drafted between two pairs of rollers alone, the way cotton is drafted?
2. During the study you watch a faller bar complete its journey. Describe the three things its pins do, in order.
3. The reach on a drawing frame is found set shorter than the cut length of the material. What is happening to the fibre, and what two symptoms would you expect in the delivered sliver?
4. A frame's retaining rollers run at 1.6 m/min and its drawing rollers at 14.4 m/min; six ends of feed sliver are doubled. What draft is applied, and what is the relation between feed and delivered sliver

weight?

5. The same count of jute yarn is ordered as warp and as weft. Which frame setting differs between the two productions, in which direction, and why?
6. A jute yarn is labelled 12 lb and a cotton yarn 12 Ne. Which is the coarser product, and what does each number actually state?

Answer key

1. Jute strands are up to metres long and highly variable; rollers set far enough apart not to break them cannot grip and meter them, and rollers set close would tear them. The pinned faller field carries the whole strand mass at a controlled speed while individual fibres are drawn through the pins — control along the whole length, which rollers cannot give.
2. (i) The pins rise into the sliver at the back of the field and take charge of the strand mass; (ii) they carry the fibres forward at retaining speed, holding back every fibre whose leading end has not yet been seized by the drawing rollers; (iii) at the front they drop away clear of the drafted sliver and return beneath the field for the next cycle.
3. Fibres are being gripped at both ends at once — seized by the drawing rollers while still held by the retaining nip and pins — so the draft tears strands instead of sliding them. Symptoms: raised short-fibre content with fly and dust at the frame, and slubs with weight variation in the delivered sliver.
4. Draft = 14.4 ÷ 1.6 = **9**. With 6 doublings, delivered weight = (6 × feed weight) ÷ 9 = two-thirds of one feed end's weight per unit length — draft and doubling must always be stated together, since either alone says nothing about the delivered sliver.
5. Twist. The warp production is set to the higher twist (≈ 3.2–3.8 tpi against ≈ 2.4–3.0 tpi for weft, adjusted in the drive gearing): warp yarn must survive sustained tension and abrasion from heddles, reed and shuttle, while weft is kept softer because it meets little stress in the cloth.
6. The numbers state opposite things. Jute 12 lb is a direct-system count: one spindle (14,400 yd) of the yarn weighs 12 lb — a heavy, coarse yarn, and a bigger number would be coarser still. Cotton 12 Ne is indirect: 12 hanks of 840 yd weigh 1 lb — a medium-coarse cotton yarn, and a bigger number would be finer. Of the two, the jute yarn at 12 lb per 14,400 yd (5,443 g over 13,167 m ≈ 0.41 g/m, about 413 tex) is far coarser than 12 Ne cotton (≈ 49 tex, 0.049 g/m).

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib